

# **CSA1717-ARTIFICIAL INTELLIGENCE**

## **PRACTICAL LAB EXPERIMENTS**

**NIHARIKA PREEJESH (192524506)**

### **Aim:**

To write a Prolog program to find the **number of vowels in a given string**.

### **Objective:**

To identify vowels (a, e, i, o, u) from a string and count their total number using recursion.

### **Algorithm:**

1. ☐ Start the program.
2. ☐ Read a string as a list of characters.
3. ☐ Take the first character.
4. ☐ Check whether the character is a vowel.
5. ☐ If it is a vowel, increment the count.
6. ☐ If it is not a vowel, continue without incrementing.
7. ☐ Recursively process the remaining characters.
8. ☐ When the list becomes empty, return the total count.
9. ☐ Stop.

## **Prolog Program:**

% Program to count vowels

vowel(a).

vowel(e).

vowel(i).

vowel(o).

vowel(u).

count\_vowels([], 0).

count\_vowels([H|T], Count) :-

    vowel(H),

    count\_vowels(T, Count1),

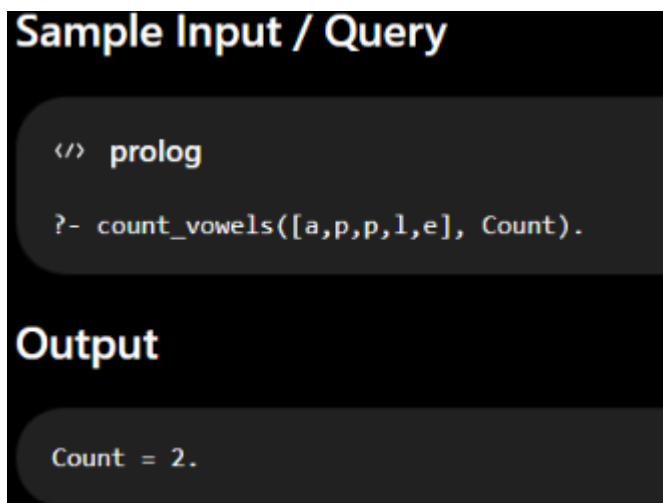
    Count is Count1 + 1.

count\_vowels([H|T], Count) :-

    \+ vowel(H),

    count\_vowels(T, Count)..

## Output

A screenshot of a Prolog interpreter interface. It features a dark background with light-colored text. The interface is divided into two main sections. The top section is titled "Sample Input / Query" in a bold, light-colored font. Below this title, there is a light gray rounded rectangle containing two lines of Prolog code: the first line is a comment starting with a Prolog comment symbol and the word "prolog", and the second line is a query starting with a question mark and a dash, followed by the predicate "count\_vowels" with two arguments: a list of characters "[a,p,p,l,e]" and a variable "Count". The bottom section is titled "Output" in a bold, light-colored font. Below this title, there is a light gray rounded rectangle containing the output text "Count = 2." in a light-colored font.

```
</> prolog
?- count_vowels([a,p,p,l,e], Count).
```

**Output**

```
Count = 2.
```

## Result:

Thus, the Monkey Banana Problem is successfully represented using Prolog.